



C23-EC-305

23133

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DEC – THIRD SEMESTER EXAMINATION

NETWORK ANALYSIS

Time : 3 hours]

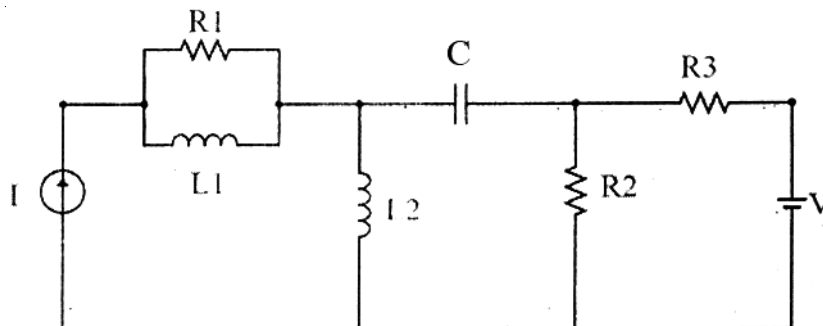
[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Determine the number of mesh equation required to solve the given network :



2. Mention the concept of duality.
3. State reciprocity theorem.
4. Give transformation formulas for delta to star.

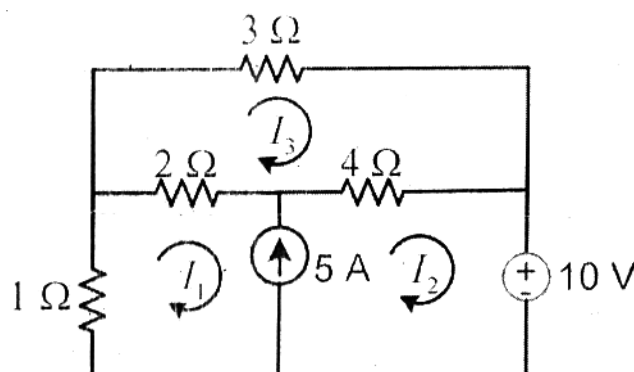
5. Define lower cut-off and upper cut-off frequencies.
6. List any three comparisons between series and parallel resonance.
7. Write Laplace transforms of (a) unit step function, (b) cosine function and (c) damped hyperbolic sine function.
8. Define the terms (a) initial condition, (b) steady state and (c) transient state.
9. Define the terms propagation constant and attenuation.
10. List the disadvantages of constant K fillers.

PART—B

10×5=50

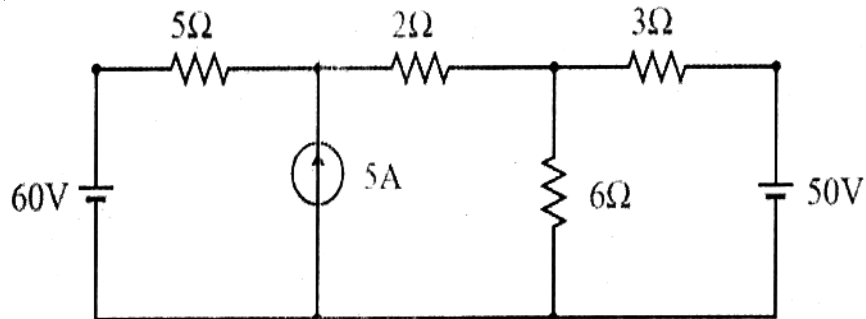
- Instructions :**
- (1) Answer *any five* questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Solve for mesh currents using Cramer's rule for the given network : 10



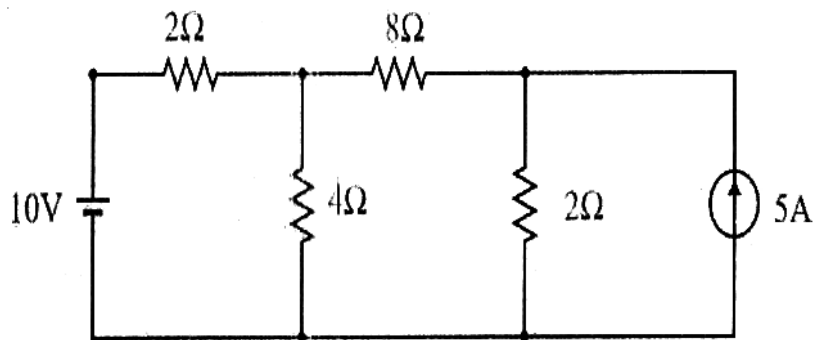
12. Find node voltages in the circuit shown in figure below using nodal analysis :

10



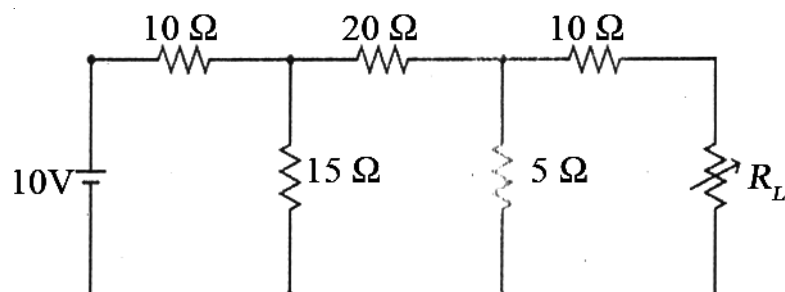
13. Find the voltage across $4\ \Omega$ resistor in the circuit shown in figure below using superposition theorem.

10

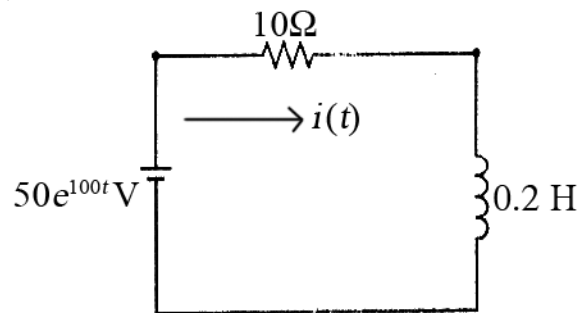


14. Determine the value of R_L and maximum power in the circuit shown in figure below for maximum power transfer :

10



- 15.** A series R - L - C circuit with $R = 10\ \Omega$, $L = 1\ \text{mH}$ and $C = 1000\ \text{pF}$ capacitor is connected across a sinusoidal source of $20\ \text{V}$ with variable frequency. Calculate the following : 10
- (a) The resonant frequency
- (b) Q -factor of the circuit at resonance frequency
- (c) Lower cutoff frequency
- (d) Upper cutoff frequency
- 16.** Explain the DC response of series R - L - C circuit. 10
- 17.** Determine the current $i(t)$ for $t \geq 0$ if initial current through the inductor $i(0) = 0$ for the series R - L circuit shown in figure below using Laplace transformation method : 10



- 18.** (a) Explain Π type attenuator with circuit diagram. 6
- (b) Design a Π type attenuator having $20\ \text{db}$ attenuation and characteristic impedance of $600\ \Omega$. 4

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